7.1 Angles of Polygons

Name _____ Period ____ Date _____

Geometry 7.1 (30-minute plan) · Student sheet

I can…

- find the sum of the interior angle measures of a polygon.
- use interior and exterior angle sums to find unknown angle measures.
- find each interior and exterior angle of a regular polygon.

Vocabulary · fill in the missing word

Diagonal — a segment that joins two _____ vertices of a polygon

Exterior angle — an angle formed by one side and the _____ of the next side

Equilateral / equiangular polygon — all sides _____ / all angles congruent

Regular polygon — a convex polygon that is both _____ and equiangular

Part 1 · Polygon names and triangles in a polygon

Polygon names (keep this list for the whole lesson)

Sides	Name	Sides	Name
3	triangle	8	octagon
4	quadrilateral	9	nonagon
5	pentagon	10	decagon
6	hexagon	12	dodecagon
7	heptagon	n	n-gon

Triangles in a polygon (from the TV): from one vertex, draw every diagonal and count the triangles.

Number of sides	Triangles from one vertex	Sum of the interior angles
4		
5		
6		
n		

Part 2 · Notes and examples (fill in each note, then work the example)

Polygon Interior Angles Theorem: The sum of the interior angle measures of a convex n-gon is _____.

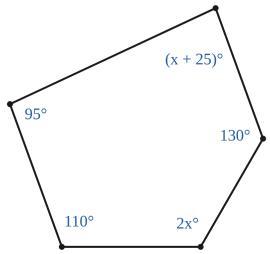
Corollary: The interior angle measures of a quadrilateral add to _____°.

Example 1

Find the sum of the interior angle measures of a convex 12-gon.

Example 2

A pentagon has interior angles of 110°, 2x°, 130°, (x + 25)°, and 95°. Find x and the two unknown angles.



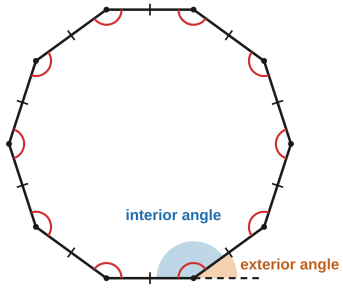
Polygon Exterior Angles Theorem: The exterior angles of a convex polygon, one at each vertex, add to _____°.

Regular n-gon: each interior angle = _____ each exterior angle = _____

Example 3

Find the measure of each interior angle and each exterior angle of a regular decagon.

regular decagon: 10 congruent sides, 10 congruent angles



Part 3 · Partner practice

Partner practice

Each exterior angle of a regular polygon measures 24°. How many sides does it have?